Design Optimization of Asymmetric Patterns for Variable Stiffness of Continuum Tubular Robots

Soyeon Park, Jongwoo Kim, *Member, IEEE*, Chunwoo Kim, *Member, IEEE*, Kyu-Jin Cho, *Member, IEEE*, and Gunwoo Noh

***Abstract*—Variable stiffness mechanisms have become popular for improving the performance of meso/microscale robots due to their ability to enable safe, effective, and versatile operation without additional actuators. In minimally invasive surgery (MIS), they can provide low stiffness for adaptability and an extended workspace and high stiffness for stable manipulation. This study aims to optimize the pattern design that maximizes the stiffness ratio while preventing buckling. Longitudinal slits are identified as the ideal shape for maximizing the ratio using topology optimization. The effects of the design parameters are investigated by analytical modeling and finite-element analysis (FEA). Finally, we established a design optimization process to maximize the variable stiffness ratio while ensuring safety. The load and buckling tests of 19 cases are verified in both the experiment and FEA. Buckling occurred in eight cases, consistent with the expected results from the proposed map. Among the tested cases, the tube with the optimal pattern parameter set exhibits the highest stiffness ratio of 2.67 while satisfying the given stiffness constraint (greater than 0.3 times the flexural stiffness of the initial tube) and avoiding buckling. The proposed optimization method potentially enables an extended workspace and more versatile functionality for MIS instruments.**

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***Index Terms*—Mechanism design, medical robots and systems, steerable catheters/needles, surgical robotics, variable stiffness mechanism.**

I. Introduction

OVER the past two decades, minimally invasive surgery (MIS) has become increasingly popular due to its advantages over traditional open surgery, such as smaller incisions, reduced blood loss, improved cosmesis, and faster recovery [1]. MIS procedures have replaced conventional methods in several surgical procedures, including heart, thoracic, neuro, orthopedic, and urologic surgery [1]–[3]. Continuum robots are widely used in MIS due to their scalability and compactness [4]. However, the smaller the operating space, the more difficult it is to control the motion of continuum robots because of their smaller moment arm and lack of micro-sized actuators. Instead of adding actuators, an alternative method of motion control is controlling stiffness. Under the same load, displacement increases as flexural stiffness decreases. Varying its stiffness can be an effective method for controlling the motion of the MIS instrument. Furthermore, low stiffness can increase the capability of the system to absorb impacts to minimize tissue damage. On the other hand, high stiffness can increase the control bandwidth and position-tracking precision for tissue manipulation and prevent buckling. This desire has led to the development of scalable and compact variable stiffness mechanisms for meso/microscale robots.

Researchers have studied diverse variable stiffness mechanisms: shape-memory materials [5], anisotropic pneumatic forces [6], variable neutral-line mechanisms [7], layer jamming [8], granular jamming [9], [10], fiber jamming [11], magnetic force control [12], low-melting-point alloys [13], polymer modulations [14], air inflation [15], and lever mechanisms [16]. However, most current designs for variable stiffness mechanisms are excessively complex to scale down to 3 mm or less in diameter. Mechanisms that use high heat or voltage stimuli introduce potential safety issues during surgical procedures. Those based on shape-memory polymers, conductive elastomers, or low-melting-point alloys have relatively long response times (> 17 s) to shift their state due to heating and cooling delays [13]. These delays are undesirable because surgeons aim to reduce operating time for patient safety and faster recovery. Moreover, many current mechanisms exhibit a discrete change of stiffness that limits suitable applications.

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TABLE I
PATTERN DESIGNS FOR CONTINUUM TUBULAR ROBOTS

	Target Function	Pattern Shape	Buckling Analysis	Parametric Study
Haga et al. [46] Bell et al. [23] York et al. [24]	High bending degree	Rectangular, 1 column†	No	Limited
Kutzer et al. [25] Liu et al. [26]	High bending degree	Rounded rectangular, 2 columns†	No	Limited
Wei et al. [27]	High bending degree	Triangular, 2 columns†	No	No
Kim et al. [28]–[30]	High curvature	Horizontal, 4 columns†	No	No
Lee et al. [18]	Reduce EI/GJ	Horizontal, 12 columns†	No	Limited
Luo et al. [21]	Reduce EI/GJ	Diamond, 10 columns†	No	No
Kim et al. [17]	Variable stiffness	Dogbone, 4 columns†	No	Limited
Current study	Variable stiffness	Rectangular, 2 columns*	Yes	Full (Automatic FEA)

General shape of the pattern was decided by physical intuition† or topology optimization∗ FEA: Finite Element Analysis.

In the previous study, we proposed a variable stiffness mechanism for controlling continuum robots in a limited space using patterned superelastic tubes [17]. We used the laser patterning method to build a continuously variable stiffness mechanism that consists of two or more coaxial tubes. The nonuniform patterning on each tube creates stiffness anisotropy. The stiffness of the proposed tubular structure with respect to a particular radial direction is controlled predominantly by relative rotation between the coaxial tubes. The mechanism differs from other variable stiffness mechanisms because it offers continuously variable stiffness, scalability, and rapid response times. A tubular robot structure was selected so its hollow space could be used for the passage of surgical tools, such as cautery probes, graspers, and forceps.

Many patterning methods on continuum tubular robots have been studied. Researchers aimed to refine material properties to solve instability problems [18]–[21], create a compliant region for bending [22]–[27], use a configuration with high-curvature [28]–[30], or use a variable stiffness mechanism [17] (see Table I). In each study, different pattern shapes (e.g., diamond or dog-bone) and distributions (e.g., the number of longitudinal columns of patterns) are applied for its applications.

The pattern designs and their analysis are compared in Table I. For example, the long horizontal slit pattern was utilized to solve the instability problem of precurved tubes [18], and the dog-bone shape was used to design the continuously variable stiffness mechanism [17]. However, those studies determined the general pattern shape based on "intuition," and parametric studies on the pattern shape were not considered or were performed by "limited sensitivity" analysis with the assumption of independence of variables.

The topology optimization approach has been widely used not only in the field of structural component design in the automotive and aerospace industries [31]–[33] but also in the fields of thin-walled structures [34] and truss structures [35]. It has become increasingly popular in a wide range of industries because of its development through a commercial finite element (FE) analysis software program and easy-to-use implementation of algorithms [32], [33], [35], [36]. In recent years, by applying topology optimization in pattern design, an initial design space can be modified for a predefined objective with associated constraints [34], [37], [38].

The topology optimization process was conducted as follows: constructing an FE model for the initial design space, generating an optimized topology after setting optimization problems, and performing postprocessing. Regarding the topology optimization of patterns on concentric tube robots, Luo *et al*. [21] attempted to reduce the bending-to-torsional-stiffness ratio to solve the snapping problem in a specific tube set. They used the solid isotropic material with the penalization (SIMP) method to determine the general shape. However, they obtained nonconverged results, which were different solutions for different setups. These results indicated that local minimum or nonunique solutions appeared in the structural topology optimizations, which was affected by the mesh dependence or local minimum problem [21], [39]–[41]. In postprocessing, the solution seems insufficient for determining the optimized pattern as a diamond shape because the final design was determined based on nonconverged results and the average dimensions of the pattern used to create a standardized repeat pattern on the tube. The parametric study was not performed to study the sensitivity of the design variables.

Furthermore, in previous studies (see Table I), the buckling analysis of patterned tubes has not been addressed. However, buckling is a common concern for slender and tubular continuum robots and may lead to unexpected motions [42]. Sudden and unexpected motions during buckling can cause the continuum robot to collide with nearby objects or break the robot itself.

An advanced parametric study is also required to cover a wide range of design parameters. In this article, we focus on optimizing pattern design and deriving its design guidelines for continuum tubular robots. We aim to

1) determine the general shape and its distribution,
2) parameterize the optimized pattern,
3) maximize the variable stiffness capacity,
4) while preventing buckling.

We proposed a new optimization process to achieve this goal.

First, we determined the general pattern shape using SIMP algorithms. The three different optimization setups were considered using different objective functions and constraints to verify the converged solution. The converged solution based on SIMP algorithms (gradient-based optimization methods) can produce a general shape without any biased shape composition. With the obtained general shape, we can parameterize design factors accordingly. We then constructed a reduced analytical parametric model based on the topology results and determined the significant parameters of the general pattern for FE parametric studies.

Second, an FE parametric study using automatic simulation was performed to "map" the performance of the tube for all

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possible parameter values. Buckling may lead to unwanted movement, fatigue failure, or breakage of a robot, thus damaging nearby objects or the robot itself. Design variable conditions to prevent bending and torsional buckling, which are the two major types of buckling of the suggested tubular continuum robots, were considered to avoid instability while optimizing performance. In this process, nonlinear buckling analyses were performed within an appropriate range of parameters to constrain the pattern parameter. Consequently, it was possible to find the optimal design variable combination from the resulting map while preventing buckling.

The contribution of the proposed design process is as follows:

1) topology optimization,
2) analytic model,
3) parametric study,
4) buckling analysis,
5) optimization procedure using all these steps.

The use of an automatic FE parametric study and mapping of a wide range of simulation results are new approaches. Furthermore, no research has considered nonlinear buckling simulations in FE parametric studies to ensure the stability of the tube. With this procedure, we can customize the pattern design parameters to achieve the desired variable stiffness ratio while satisfying the given constraints chosen for the loading conditions for surgical use and minimum compliance for securing the desired tube performance.

II. Design Optimization of Mechanism

A. Goal of Pattern Design Optimization

The proposed mechanism consists of multiple concentric tubes with an anisotropically distributed pattern. The patterns of each of the tubes are geometrically similar. Due to the anisotropic pattern, the rotational configuration of the tubes determines the stiffness of the mechanism along a specific direction. We maximize the variable stiffness ratio by seeking a pattern design that provides a large stiffness ratio between two mutually perpendicular directions. For example, the total stiffness I_{total} of the two-tube set (inner tube and outer tube) is the linear summation of the stiffness of each tube:

$$I_{total,xx} = I_{inner,xx} + I_{outer,xx} \tag{1}$$

$$I_{total,yy} = I_{inner,yy} + I_{outer,yy}. \tag{2}$$

Because the inner and outer tubes have the same anisotropic distribution along the radial direction, min $(I_{total,yy}/I_{total,xx})$ is determined to find min $(I_{inner,yy}/I_{inner,xx})$ and min $(I_{outer,yy}/I_{outer,xx})$ and, consequently, to find min (I_{yy}/I_{xx}) for the single tube. Thus, the problem is reduced to finding min (I_{yy}/I_{xx}) for a single tube.

We studied an optimal cutting pattern to obtain the least values of I_{yy}/I_{xx} within the constraints. We first identified the general shape of the through-hole pattern by performing topology optimization. The parameters for the general pattern shape and longitudinal slits were quantified for the designated minimum ratio by a reduced analytical parametric model of the structure. Subsequently, the parameter was optimized by performing an FE parametric study. A nonlinear buckling analysis was performed to provide the boundaries of the design parameters to ensure the stability of the tube mechanism.

B. Topology Optimization for Pattern Shape

We identified an optimized pattern without using a predetermined shape by performing a topology optimization using the SIMP method. SIMP-based topology optimization is a gradient-based optimization method that changes the density of each element in each iteration to minimize the objective function [39]. The user defines a meshed design space in which the density of each element ranges between 0 and 1, where 0 is fully void and 1 is fully dense. Each iteration of the algorithm then adjusts the density of individual elements based on sensitivities and the surrounding elements. It optimizes the material layout within a given design space for a given set of loads, boundary conditions, and constraints to maximize system performance. We used the TOSCA structure and ABAQUS 6.14 (Dassault Systems) for topology optimization. The general pattern shape was analyzed to maximize its variable stiffness ratio while satisfying the given constraints.

For topology optimization, we considered a unit-patterned short tube with an outer diameter (OD) of 2.03 mm, an inner diameter (ID) of 1.33 mm, and a length of 4 mm. For minimizing the stiffness ratio, I_{yy}/I_{xx}, the angles of the tube at the reference points were used as the design variables. The resulting angle of the tube, *r*, was measured at the distal tip of the tube, which is denoted by r_x and r_y based on the directions of *x* and *y*, respectively. According to elementary beam theory, the angle of the tube, r, is inversely proportional to stiffness under the same moment, Young's modulus, and length of the tube. The relational expression for the bending moment, *M*, and resulting angle of the tube, r, is $M = EI(r/L)$, where EI is the flexural rigidity and *L* is the total length of the tube.

Three different optimization setups were tested using different objective functions and constraints to verify the converged solution. The objective functions min (r_x/r_y), min (r_x), and max (r_y) were used in Setups 1, 2, and 3, respectively. In Setup 1, the objective function is set to maximize the *x*-axis stiffness and minimize the *y*-axis stiffness, simultaneously. Also, in Setup 1, the angle of the distal tip, r_y, was constrained to be more than 50% of initial r_y. In Setup 2, the objective function used for maximizing the *x*-axis stiffness while $r_y$was constrained to be more than 50% of initial r_y. In Setup 3, the objective function was used for minimizing the *y*-axis stiffness while $r_x$was constrained to be less than two times of initial r_x. In all the setups, the volume was constrained to be reduced by up to 30% compared to the initial volume of the tube. A geometric restriction was also set to avoid thin trusses that are difficult to manufacture: we specified the minimum size of the member as 0.1 mm.

The optimization results are shown in Fig. 1. The high-density areas are indicated in red. As the density decreases, the color progressively changes to blue. In all cases, the distribution of the low-density areas converges to the shape of a longitudinal

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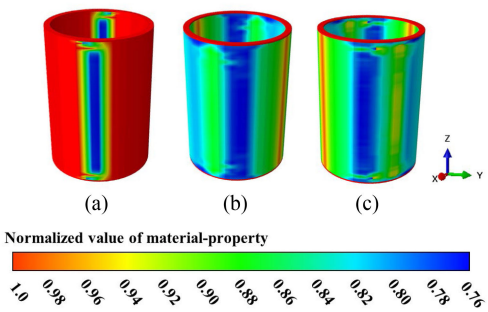

Fig 1. Topology optimization results from (a) Setup 1, (b) Setup 2, and (c) Setup 3.

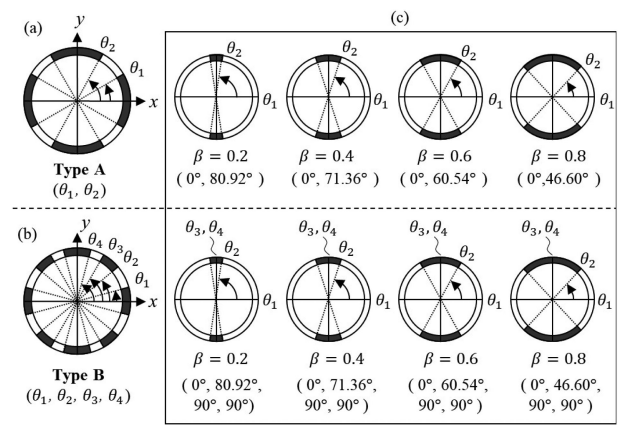

Fig. 2. Cross section of the patterned tube. (a) Type A has one slits in each quadrant and (b) Type B has two slits in each quadrant. (c) Each case demonstrates the optimized design parameters, θ_1, θ_2, θ_3, and θ_4, to maximize γ $(= I_{xx}/I_{yy})$ under the given constraint, $0.2 \leq \beta(= I_{xx}/I_{xx0}) \leq 0.8$ where I_{xx0} is I_{xx} of the nonpatterned tube. For all cases, θ_1 is zero and θ_3, θ_4 are 90 °. Both types A and B converge to the same design when the same β is given.

slit. In the topology results, the stiffness ratios (I_{yy}/I_{xx}) are 0.58 at Setup 1, 0.35 at Setup 2, and 0.38 at Setup 3. The optimized results were consistent enough to determine the general shape, and the "longitudinal slit" was chosen as the pattern design. The patterning decreased the flexural stiffness (EI) of the unit, and it also led to an anisotropic distribution of flexural stiffness in the cross sections.

C. Analytical Modeling for Pattern Design

The stiffness of a tube can be modeled as the sum of the stiffness of each section of the tube divided by a specific central angle θ_i. Each section has the second moment of area I, which is the dominant factor in determining the flexural stiffness of the mechanism. We considered that the patterned section has zero flexural stiffness because of the lack of any material. Assuming that a tube is divided into a patterned section $(E_p = 0)$ and nonpatterned section $(E_u = E)$, the flexural stiffness of the tube can be modeled as the sum of the stiffness of all sections, where i is the number of sections, as depicted in the following equation:

$$(EI)_{tube} = \sum (E_p I)_i + \sum (E_u I)_i = \sum (EI)_i. \quad (3)$$

The stiffness of the mechanism is determined by the pattern design parameters, which are the geometric properties of the tube. These geometric properties consist of the height and width of the slit, which determines the stiffness range. Slit width depends on the angle from the center with respect to the cross section, and slit length determines the length of the unit cell of the tube. Another design variable is the number of patterns in one tube unit.

Two types of patterns, A and B, as shown in Fig. 2(a) and (b) were created on the tubes for comparison. The white section is patterned (through-hole), whereas the black section is nonpatterned. Type-A and -B patterns have one and two slits in each quadrant, respectively. Because the Type-A pattern has one slit in one quadrant, the second moment of the area with the Type-A pattern can be expressed as follows using θ_1 and θ_2 in the range of 0 rad to $\frac{\pi}{2}$ rad

$$I_{xx} = \int_{\frac{d_i}{2}}^{\frac{d_o}{2}} \left[\int_0^{\theta_1} (r\sin\theta)^2 + \int_{\theta_2}^{\frac{\pi}{2}} (r\sin\theta)^2 \right] r d\theta dr$$
$$= \frac{d_o^4 - d_i^4}{64} \left[\int_0^{\theta_1} (\sin\theta)^2 d\theta + \int_{\theta_2}^{\frac{\pi}{2}} (\sin\theta)^2 d\theta \right] \quad (4)$$
$$I_{yy} = \frac{d_o^4 - d_i^4}{64} \left[\int_0^{\theta_1} (\cos\theta)^2 d\theta + \int_{\theta_2}^{\frac{\pi}{2}} (\cos\theta)^2 d\theta \right]. \quad (5)$$

With (4) and (5), we obtain the stiffness ratio between x- and y-axes as follows:

$$\frac{I_{yy}}{I_{xx}} = \frac{\theta_1 - \theta_2 + \frac{1}{2}\sin(2\theta_1) - \frac{1}{2}\sin(2\theta_2) + \frac{\pi}{2}}{\theta_1 - \theta_2 - \frac{1}{2}\sin(2\theta_1) + \frac{1}{2}\sin(2\theta_2) + \frac{\pi}{2}}. \quad (6)$$

We determined the number of vertical slits by considering two longitudinal slits in one quadrant, shown as the Type-B pattern in Fig. 2. The second moment of the area for this pattern can be expressed as follows using θ_1, θ_2, θ_3, and θ_4 in the range of 0 rad to $\frac{\pi}{2}$ rad

$$I_{xx} = \frac{d_o^4 - d_i^4}{64} \times \left[\int_0^{\theta_1} (\sin\theta)^2 d\theta + \int_{\theta_2}^{\theta_3} (\sin\theta)^2 d\theta + \int_{\theta_4}^{\frac{\pi}{2}} (\sin\theta)^2 d\theta \right] \quad (7)$$

$$I_{yy} = \frac{d_o^4 - d_i^4}{64} \times \left[\int_0^{\theta_1} (\cos\theta)^2 d\theta + \int_{\theta_2}^{\theta_3} (\cos\theta)^2 d\theta + \int_{\theta_4}^{\frac{\pi}{2}} (\cos\theta)^2 d\theta \right]. \quad (8)$$

With Equations (7) and (8), we obtain the stiffness ratio (I_{yy}/I_{xx}) between the two directions of the tube with the Type-B pattern as, eq. (9) as shown at the bottom of the next page.

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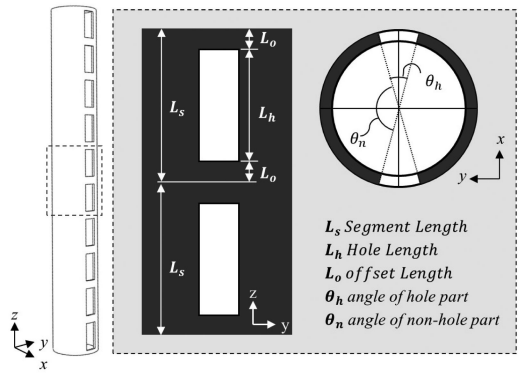

Fig. 3. Design parameters for optimized pattern shape.

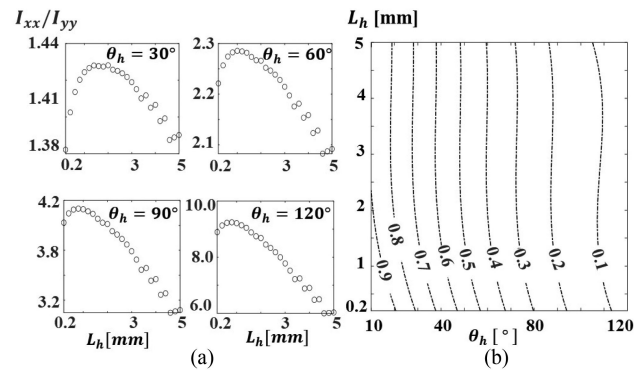

Fig. 4. (a) Map of $\gamma(= I_{xx}/I_{yy})$ to find the hole length that maximizes γ when $\theta_h \in (30°,\ 60°,\ 90°,\ \text{and } 120°)$ for stainless-steel tube. (b) Map of β for stainless-steel tube.

As shown in (6) and (9), the second moment of the area of the section is determined by the hole angles. For both pattern types, we calculated the hole angles resulting in the minimum of I_{yy}/I_{xx} using the multivariate optimization given in the following equation:

$$\begin{aligned} &\text{Objective function}: \ \min. f(\theta) = \frac{I_{yy}(\theta)}{I_{xx}(\theta)} \\ &\text{subjected to}: \\ &I_{xx} \geq \beta \cdot I_{xx0},\ I_{yy} < I_{xx} \\ &\pi/2 \geq \theta_4 \geq \theta_3 \geq \theta_2 \geq \theta_1 \geq 0 \end{aligned} \tag{10}$$

where $\beta(= I_{xx}/I_{xx,0})$ is used to indicate the minimum compliance of the structure, and I_{xx0} is I_{xx} of the nonpatterned tube. The constraint $I_{xx} \geq \beta \cdot I_{xx0}$ is given for $0.2 \leq \beta \leq 0.8$.

The optimized results and cross sections for the Type-A and -B patterns are shown in Fig. 2(c). In Fig. 2, we express the objective function as the reciprocal $\gamma(= I_{xx}\ /I_{yy})$ for convenience. For maximizing γ, the optimized Type-A patterns have a zero value of θ_1 and a nonzero value of θ_2. In all cases of optimized Type-B pattern, θ_1 is zero, θ_3 and θ_4 are 90°, and θ_2 has a non-zero value. In both types, the nonzero angle is equally determined by the constraint β, and both the Type-A and -B patterns have a pair of longitudinal slits that are symmetrical to the origin. Even if there are several pairs of longitudinal slits in the initial state, the optimized pattern converges on one pair of longitudinal slits.

D. Parametric Study of Pattern Design Parameters

Based on the results of the general shape and number of patterns, the following design parameters for unit cell patterning were determined: L_s(segment length of the unit cell), L_o(offset length of the unit cell), L_h(hole length of the unit cell), θ_h(angle of slit part), and θ_n(angle of nonslit part), where $\theta_h + \theta_n = \pi$ and $L_s = 2L_o + L_h$ (see Fig. 3). We maximize the ratio $\gamma(= I_{xx}\ /I_{yy})$ by setting L_o to 0.25 mm, which was relatively small compared to *Ls*. The stiffness range is determined by the pattern design parameters. L_h and θ_h denote the height and angle of a unit pattern, respectively. The results demonstrate that θ_h is a dominant design parameter in determining the stiffness ratio (I_{yy}/I_{xx}), while L_h is critical for avoiding buckling, as discussed in the following section.

TABLE II
SPECIFICATIONS OF NITINOL AND STAINLESS-STEEL TUBES

Material		$d_i(mm)$	$d_o(mm)$	$E(GPa)$
Nitinol	Inner tube	1.8	2.0	67.6
	Outer tube	2.2	2.4	51.3
Stainless-steel	Inner tube	1.51	1.83	188

The stiffness ratio, which varies with the θ_h and L_h of the pattern, was evaluated by the FE simulation of the load test. The test was performed for $\theta_h \in (10°,\ 15°,\ 20°,\ \ldots 120°)$ and $L_h \in (0.2$ mm, 0.4 mm, 0.6 mm, …5 mm); a total of 575 cases were tested per specimen. In the simulation, the load was selected to be less than 400 g and was applied at the distal end of the mechanism. For calculating bending stiffness (*EI*), vertical deflections along the *x*- and *y*-axes of the distal tip were measured to calculate the stiffness ratio.

As the length of the hole changes from 0 to 5 mm, there is a maximum point that indicates the hole length L_h to be used to maximize γ at a specific θ_h. In Fig. 4(a), the results of the four representative cases $\theta_h \in (30°, 60°, 90°,$ and $120°)$ among the various angles are presented for the stainless-steel tube in Table II. The cases, $(\theta_h,\ L_h,\ \gamma)$, where the maximum value of γ appears, are as follows: (30°, 1.2 mm, 1.43), (60 °, 0.8 mm, 2.28), (90 °, 0.6 mm, 4.13), and (120°, 0.6 mm, 9.23).

We also evaluated the values of $\beta = (I_{xx}\ /I_{xx0})$ to ensure minimum compliance for securing the desired tube performance. Fig. 4(b) illustrates the values of β for various θ_h and L_h. The results demonstrate that the value of θ_h had the dominant effect in terms of determining the minimum stiffness of the tube, which

$$I_{yy}/I_{xx} = \frac{\theta_1 - \theta_2 + \theta_3 - \theta_4 + \frac{1}{2}\left[\sin(2\theta_1) - \sin(2\theta_2) + \sin(2\theta_3) - \sin(2\theta_4) + \pi\right]}{\theta_1 - \theta_2 + \theta_3 - \theta_4 - \frac{1}{2}\left[\sin(2\theta_1) - \sin(2\theta_2) + \sin(2\theta_3) - \sin(2\theta_4) - \pi\right]} \tag{9}$$

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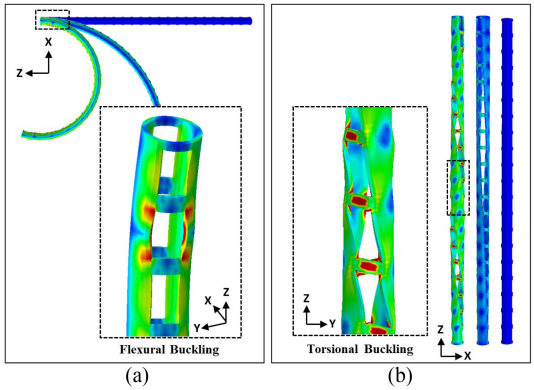

(a) (b)

Fig. 5. (a) Flexural buckling induced by bending moment (M_y). (b) Shear buckling induced by torque (M_z).

can be indicated by the value of β, whereas L_h had almost no effect.

III. Analysis of Buckling Conditions

A. Conditions of Design Parameters for Stability

The proposed patterned structure can buckle through two major routes: One is flexural buckling caused by the bending moment M_y and the other is torsional buckling caused by twist moment M_z, as illustrated in Fig. 5. A nonlinear buckling analysis was performed to limit the design parameter and ensure the stability of the variable stiffness mechanism. The nonlinear analysis was performed using the Riks algorithm [44] in the Abaqus 6.14 software environment for predicting buckling, postbuckling, and collapse of specific structures under various loading conditions. In the buckling simulations, M_y and M_z were incrementally increased until the energy suddenly dropped to find the buckling thresholds.

For determining the buckling points, the critical points were evaluated in the load–displacement curve as the criterion for determining the onset of instability and the point of buckling; the critical points are at which the slope of the load–displacement curve changes (decreases) to zero. As the force and displacement increase, large deformation occurs, and the curve slope decreases. This process causes the structure to lose its stiffness and become unstable, resulting in buckling at the critical point.

The study was performed for $\theta_h \in$ (10°, 15°, 20°, …120°) and $L_h \in$ (0.2 mm, 0.4 mm, 0.6 mm, … 5.0 mm); a total of 575 cases were examined per specimen. Due to the limitation of the fabrication workspace, the total length of the tube is less than 50 mm. The material properties and dimensions of the tube set are listed in Table II. We considered nitinol and stainless-steel as possible materials: in many biomedical applications, both are widely used due to their mechanical properties and biocompatibility. For the nitinol tubes, the inner and outer tubes have different properties, and the stainless-steel tubes have the same material properties as the inner and outer tubes.

B. Buckling Map for Stability

In Fig. 6, the changes in the buckling thresholds of the nitinol tubes (inner/outer tube) and the stainless-steel tube (inner tube) were plotted; the curves of the maximum value of $\gamma(\ = I_{xx}\ /I_{yy})$ were also plotted in the buckling map. As shown in Fig. 6, the stability of the structure is greatly affected by L_h. In practical applications, the maximum loads should be lower than the critical buckling moments. With the buckling thresholds, we may find a suitable L_h for the two load conditions ($M_y,\ M_z$) for the given θ_h. With the maps showing the buckling thresholds, the stable and unstable areas for different values of the two loading conditions can be identified. With the curve of the maximum value of γ, it is possible to estimate the θ_h and L_h that maximize γ within the stable range. Within the range of θ_h from 10° to 120°, the critical point of L_h for maximizing γ does not exceed 2 mm.

IV. Pattern Design Process

The design parameters, hole angle θ_h and hole length L_h, should be determined to maximize the variable stiffness capability or to maximize $\gamma(= I_{xx}\ /I_{yy})$. They should also be selected to prevent the buckling of the structure under the expected loading conditions ($M_y,\ M_z$) and to maintain the minimum compliance of the structure, as indicated by $\beta(= I_{xx}/I_{xx,0})$.

As shown in Fig. 4(a), γ increases as θ_h increases while there is a critical value for L_h to maximize γ, and the effect of varying θ_h is dominant. Furthermore, the constraint β limits the maximum value of the θ_h while the possible range of L_h is hardly affected, as shown in Fig. 4(b). Therefore, we first select the largest value of the θ_h as limited by β to maximize γ. In the case where β boundary is in the unstable region determined by loading constraints θ_h is determined by the load condition boundary, see Fig. 6(c). After selecting θ_h, we select L_h to prevent buckling of the structure and further maximize γ.

In Fig. 7, we illustrate the selection process using specific examples for the map of the stainless-steel tube. In CASE 1 with $\beta_{0.8}$, we considered a case in which a portion of the maximum ratio of γ was in the stable region for the two loading conditions, $M_{y,1200}$ and $M_{z,1300}$. First, the range of θ_h was determined using $\beta_{0.8}$. In the range satisfying $\beta_{0.8}$, the stable area is marked in orange. Subsequently, the case (22°, 1.65 mm) in which the $M_{y,1200}$ boundary and $\beta_{0.8}$ line overlapped, was selected.

In CASE 2 with $\beta_{0.6}$, if the boundary of torsional buckling, $M_{z,1300}$, was below the boundary of flexural buckling, $M_{y,800}$, the case (43°, 0.75 mm) was selected, in which the $\beta_{0.6}$ boundary and top of the load condition boundary $M_{z,1300}$ intersected in the stable area; the stable area is marked in green.

In another case, as shown in CASE 3 with $\beta_{0.3}$, if the boundary of the maximum ratio of γ was below the buckling limits of $M_{y,400}$ and $M_{z,300}$, the case (75 °, 0.8 mm) in the intersection of γ and the $\beta_{0.3}$ was selected because the stiffness ratio increased as it approached the γ boundary.

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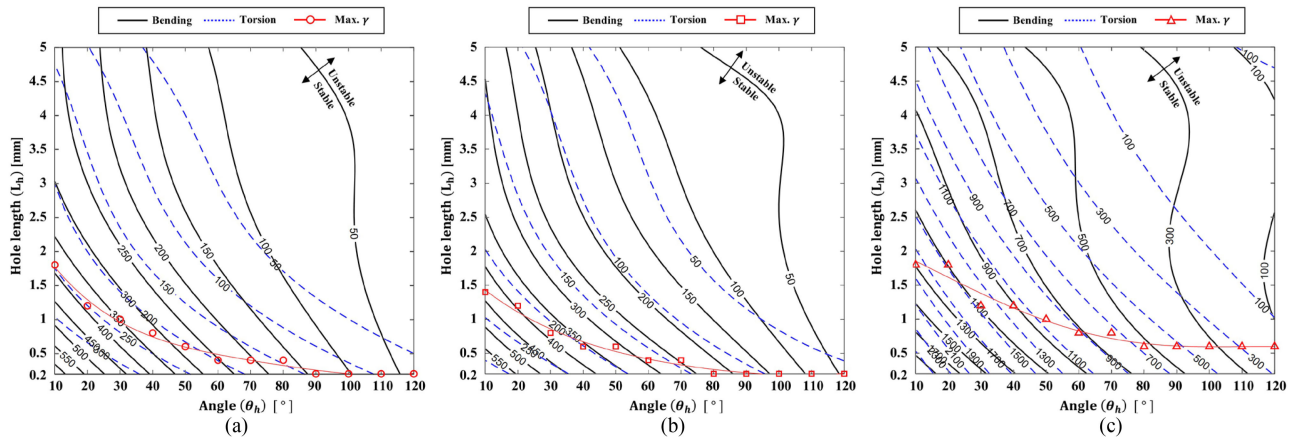

Fig. 6. Stable and unstable regions of θ_h and L_h for various loading constraints; red line indicates the hole length L_h resulting the maximum value of γ for various θ_h. The three maps represent (a) inner nitinol tube, (b) outer nitinol tube, and (c) inner stainless-steel tube.

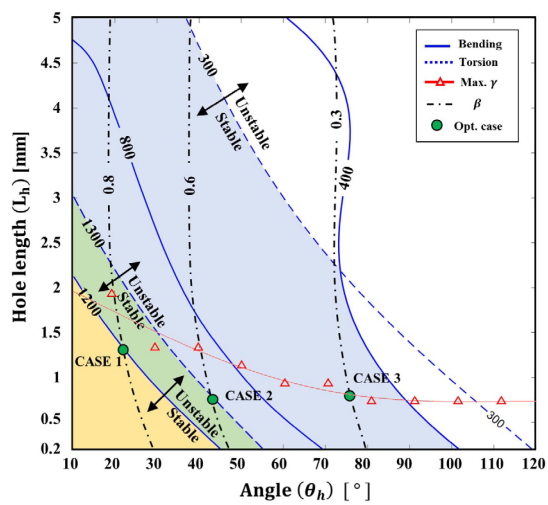

Fig. 7. For stainless-steel tube, optimal cases (θ_h, L_h, γ) in stable areas of according to maximum value of the $\gamma(=\mathrm{I_{xx}/I_{yy}})$ and buckling maps; CASE 1: (22°, 1.65 mm, 1.29) with $M_{y,1200}$, $M_{z,1300}$, $\beta_{0.8}$ in stable areas (orange area); CASE 2 : (75°, 0.8 mm, 3.02) with $M_{y,800}$, $M_{z,1300}$, $\beta_{0.6}$ in stable areas (green area); CASE 3: (75 °, 0.8 mm, 3.02) with $M_{y,400}$, $M_{z,300}$, $\beta_{0.3}$ in stable areas (blue area).

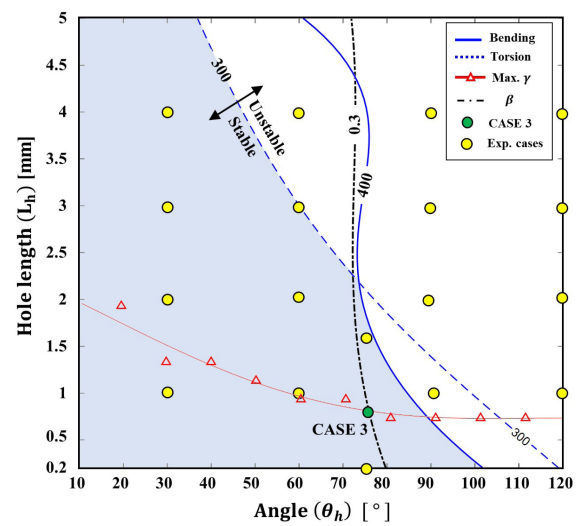

Fig. 8. For stainless-steel tube, stable areas (blue area) of θ_h and L_h according to critical moment boundary ($M_{c,y}$= 400, $M_{c,z}$ = 300) with the maximum value of the $\gamma(=I_{xx}/I_{yy})$ (red line). Yellow point indicates experimental cases to compare with map, ans green point is optimal CASE 3 determined according to the buckle map.

V. Fabrication and Experiment

The results obtained in our analysis of variations in flexural stiffness (see Section II) were verified against experimental results to confirm the stiffness variability of the mechanism and the feasibility of the model. Through-hole patterns were engraved on the stainless-steel tubes by picosecond ultraviolet (UV) laser machining. The analytical modeling and simulation results were compared with the load test results.

In the map of Fig. 8, the optimal case in the proposed map was compared for 18 experimental cases based on the buckle boundaries and curve of maximum γ to experimentally verify that the proposed design process can maximize the stiffness ratio while preventing buckling. We performed a load and buckling test, and the stiffness ratio and buckling results of CASE 3 were compared with the 18 experimental cases in this section.

A. Laser Machining for Patterning

In the proposed mechanism, the tubes have anisotropically distributed patterns of longitudinal slits to achieve variable stiffness. The designs can be scaled down to tens of micrometers, so an accurate manufacturing process is required to fabricate them. As illustrated in Fig. 9(a), a UV laser system was used for engraving the patterns on the stainless-steel tubes. For precise manufacturing, we used the picosecond UV laser system with Nd: YVO4 source (AO Pico 355 Series, Advanced Optowave, USA). Picosecond pulsewidth UV lasers have higher intensities and smaller heat-affected zones than nanosecond pulsewidth lasers.

A rotary motor stage (Unice E-O Service Inc., Taiwan) was used to hold a tube specimen as the laser scanner controlled the laser spot, as illustrated in Fig. 9(a). The rotary motor stage

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TABLE III
STIFFNESS RATIO VALUES AND BUCKLING RESULTS OF THE STAINLESS-STEEL TUBE OBTAINED IN FEA AND EXPERIMENT

θ_h (°)		30				60				75			90				120			
L_h (*mm*)		1	2	3	4	1	2	3	4	0.2	0.8	1.6	1	2	3	4	1	2	3	4
L (mm)		49.5	50.0	49.0	49.5	49.5	50	49	49.5	49.7	49.4	50.4	49.5	50.0	49.0	49.5	49.5	50.0	49.0	49.5
n		33	20	14	11	33	20	14	11	71	38	24	33	20	14	11	33	20	14	11
β		0.72	0.68	0.68	0.68	0.43	0.4	0.4	0.4	0.34	0.3	0.28	0.19	0.19	0.19	0.19	0.06	0.06	0.06	0.06
γ	*Exp*	1.44	1.50	1.36	1.36	2.23	2.25	2.10	1.96	2.61	2.67	2.59	4.36	4.14	3.29	3.19	8.80	8.12	6.77	5.96
	FEA	1.42	1.43	1.42	1.41	2.29	2.27	2.21	2.16	2.93	3.02	2.98	4.13	4.01	3.79	3.57	9.21	8.68	7.78	6.91
	Diff. (%)	1.10	4.80	4.34	3.43	2.49	0.71	5.46	10.14	12.26	13.11	15.06	5.29	3.13	15.27	11.87	4.69	6.95	14.92	15.89
F_b	*Exp*|*FEA*	X | X	X | X	X | X	X | X	X | X	X | X	X | X	O | O	X | X	X | X	X | X	X | X	O | O	O | O	O | O	O | O	O | O	O | O	O | O
T_b	*Exp*|*FEA*	X | X	X | X	X | X	X | X	X | X	X | X	X | X	X | X	X | X	X | X	X | X	O | O	O | O	O | O	O | O	O | O	O | O	O | O	O | O

Exp = Experiment, *FEA* = Finite element analysis, *Diff.* = Difference, L = Total length of specimen, n = Number of patterns in specimen along axial direction, F_b=Flexural buckling, T_b = Torsional buckling, O = Flexural/Torsional buckling occurs, X= Flexural/Torsional buckling does not occur.

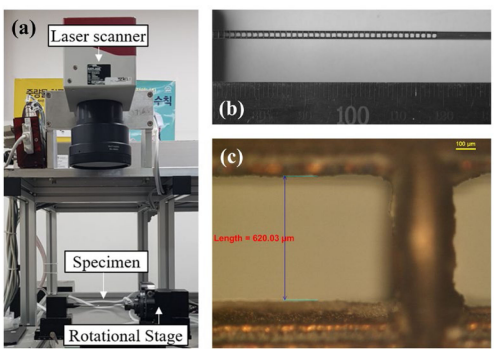

Fig. 9. (a) Picosecond UV laser system setup for patterning tubes. (b) Image of patterned tube taken using a monochrome sensor with telescopic lens when $\theta_h = 120°$ and L_h= 1 mm. (c) Microscopic photograph of engraved surface of tube when $\theta_h = 60°$ and L_h= 2 mm.

rotated the tube, which was fixed using a collet chuck. A linear guide was used to set various tube lengths, and a cone-shaped bearing support was used to prevent bending and twisting of the tube. Moreover, a glass jig was used to hold the tube specimen to achieve more precise alignment. The tubes were engraved using a pulse with a width of 7 ps, a wavelength of 355 nm, a repetition rate of 15 kHz, scanner speed of 100 mm/s, and 30 path repetitions.

After every repetition, a cooling time of 60 s was provided to minimize the effect of heat near the laser-engraving spots. The desired pattern, illustrated in Fig. 3, spanned the surface of the tube along the radial and axial directions. In this study, the surface of the tube was divided into six equal regions. Each region was patterned individually to simplify the manufacturing procedure while minimizing the error caused by the curvature of the tube surface. The surface can be divided into smaller regions based on the pattern design.

Under the effect of heat, the pattern engraved by the laser is slightly larger than the designed size. Twenty tests confirmed that the pattern length increased by an average of 15 μm compared to the design value. Therefore, we designed the pattern to be 15 μm shorter than the design value to compensate for the fabrication error and obtain more accurate patterns.

Fig. 9(b) illustrates an image of the patterned tube for θ_h = 120 ° and L_h= 1 mm. Fig. 9(c) demonstrates the microscopic photograph of the engraved tube surface. Fig. 9(b) illustrates

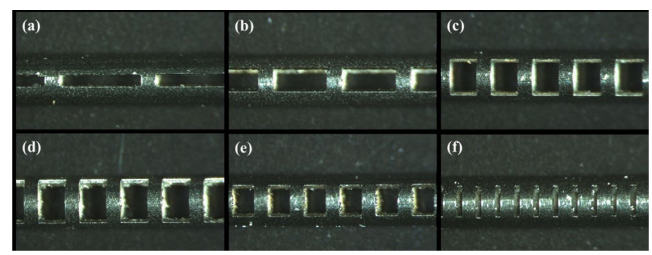

Fig. 10. Laser-machined tubes. (a) $\theta_h = 30°$ and L_h= 3 mm. (b) $\theta_h = 60°$ and L_h= 2 mm. (c) $\theta_h = 90°$ and L_h= 1 mm. (d) $\theta_h = 120°$ and L_h= 1 mm. (e) $\theta_h = 75°$ and L_h= 0.8 mm. (f) $\theta_h = 75°$ and L_h = 0.2 mm.

images captured using a telecentric lens (0.09X ½" GoldTL^{TM} Telecentric Lens, Edmund Optics, NJ, USA) coupled with a monochrome image sensor (EO-1312M, Edmund Optics, NJ, USA). Telecentric lenses minimize image distortion, and monochrome image sensors are suitable for obtaining high-resolution images of small specimens.

We validated the proposed design process and FE analysis result, illustrated in Table III and Fig. 8, by fabricating 19 specimens for $\theta_h \in (30°, 60°, 90°, 120°)$ and $L_h \in (1$ mm, 2 mm, 3 mm, 4 mm); $L_h \in (0.2$ mm, 0.8 mm, 1.6 mm) when $\theta_h =$ 75°; L_o is 0.25 mm for all cases. The number of patterns is set as Table III for a specimen length close to 50 mm. Fig. 10 depicts the microscopic photos of the selected laser machined tubes.

Despite the minimization of the heat zone effect using a picosecond UV laser system, the manufacturing system used in this study has several limitations. Due to the workspace of the laser scanner, we used specimens with lengths lesser than 50 mm. Moreover, we divided the surface of the tube into six equal regions and engraved each region sequentially. In advanced manufacturing systems, rotational and translational movements of the tube can be automatically synchronized to facilitate continuous patterning along the perimeter of the tube.

B. Load Test and Buckling Test

We compared the optimized design to other cases in terms of stiffness ratio and flexural/torsional buckling. We experimentally verified the stiffness variation ratio of the mechanism by performing a load test to compare the ratio of deflection along

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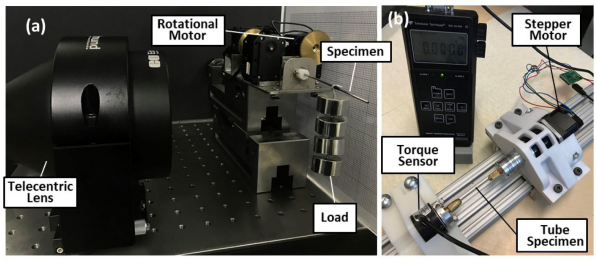

Fig. 11. (a) Load test for measuring the ratio of high stiffness to low stiffness of the variable stiffness mechanism. The rotational motor rotates the tube by 90°. Deflection of the structure is measured using the monochrome image sensor with telecentric lens. (b) Tube specimen is held by the chuck at one end and rotated by the stepper motor. The torque is measured by the torque sensor at the other end.

the two directions for the rotation angles of 0° and 90°. We completed the load test with the 19 stainless-steel specimens fabricated in Section V-A. The load was applied to the distal tip of the tube where the pattern begins while the proximal end was fixed. The deflection of the distal tip was measured. Fig. 11(a) illustrates the setup used for the load test and the flexural buckling test. The rotational motor (Dynamixel 28AT, Robotis, Korea) rotated the specimen by 90°. The deflection of the structure was measured by the monochrome image sensor coupled with the telecentric lens. Deflection δ is inversely proportional to stiffness under the same load. Thus, the deflection ratio between the two directions, δ_{yy}/δ_{xx}, had the same value as that of $\gamma(= I_{xx}\ /I_{yy})$.

We compared the deflection ratios obtained in the FE analysis with those obtained in the load test. Table III presents the ratios according to the design parameters. The average absolute difference between the simulation and experimental results is 7.94%, with a standard deviation of 5.28, demonstrating consistency with the simulation results. The difference is discussed in Section VI.

Based on Table III and Fig. 10, both in the experiment and finite-element analysis (FEA), the tube with the pattern of CASE 3 (75°, 0.8 mm) exhibited the highest stiffness ratio while satisfying the given $\beta_{0.3}$ and buckling conditions. Moreover, the stiffness ratio decreased as it moved away from the boundary of the maximum ratio of γ and boundary of $\beta_{0.3}$. Furthermore, the effect of varying $\beta_{0.3}$ was more dominant than the maximum ratio γ in both the experiment and FE analysis, which is consistent with the results of the proposed map.

The flexural and torsional buckling tests were performed to verify the simulation results. The flexural moment of 400 N·mm and the torsional moment of 300 N·mm were applied to every specimen. Fig. 11(a) and (b) demonstrates the flexural and torsional buckling test setup, respectively. The tube specimen was held by the collet chuck at one end and rotated by the stepper motor (NEMA-17 bipolar stepper motor, China). The torque was measured by the torque sensor (TRT-50, Transducer Techniques, CA, USA) on the other end.

As illustrated in Fig. 8, the conditions to prevent buckling are given with the threshold lines. In Fig. 8, the solid blue line divides the stable and unstable region for the flexural moment of 400 N·mm; the blue dashed line for 300 N·mm torsional moments. Table III compares the simulation and experimental results for buckling tests. Out of the 19 cases, buckling occurs for eight. We verified the consistency between the FEA simulation and experimental results. According to the results, the tube with $(\theta_h, L_h) = (75°, 0.8\text{ mm})$ exhibits the highest stiffness ratio while satisfying the given β stiffness and buckling conditions.

Furthermore, the proposed method achieved 8.80 for the ratio of high stiffness to low stiffness of the variable stiffness mechanism for the tube with $L_h = 1$ mm, and $\theta_h = 120°$, which is higher than the maximum value for the ratio, 4.38, from previous studies under a similar condition [17]. The proposed method also considers design to prevent buckling while the existing one does not.

VI. Discussion

In this study, we proposed a new design optimization procedure to maximize the ratio of high stiffness to low stiffness of the variable stiffness mechanism while satisfying buckling constraints. We evaluated the use of topology optimization as a starting point for pattern design. The general pattern shape obtained using the SIMP method and analytical modeling has a pair of longitudinal slits. We studied the effects of the design parameters on the variable stiffness ratio by performing automatic FEA. With the proposed design process, the maximum stiffness ratio can be adjusted to 8.80, whereas the ratio in previous studies is 4.38 [21]. The load and buckling test validated the variable stiffness ratio and flexural/torsional buckling trends according to the design parameters.

The proposed design process is summarized as follows:

1) topology optimization,
2) analytic model,
3) parametric study,
4) buckling analysis,
5) optimization procedure using all these steps.

Based on the map, one can choose the optimized design parameters according to given situations. It has advantages over the previous studies performed by "limited sensitivity" analysis with the assumption of independence of variables. In our future research, various methods will be included to further develop each process. We plan to construct the surrogate model to evaluate the performance of the concentric tube effectively. Moreover, in the step of determining the general shape of patterned tube, we will apply various optimization methods to improve the search for optimal solutions using nongradient optimization methods such as heuristic algorithms.

We investigated the variable stiffness capabilities to overcome the limitations of current bending wrists. The proposed variable stiffness mechanism can be used to control the motion of continuum instruments for MIS. The proposed optimized pattern can reduce the unreachable area of surgical instruments by controlling their stiffness. The extended variable stiffness range can broaden the workspace of MIS instruments, such as steerable needles/cannulas and concentric tube robots. Based on the proposed buckling maps, we can design our continuum robots to avoid executing undesired motions.

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At the micro-scale, laser-cutting edges cannot be flat and clean, as assumed in the simulation. The cutting edges should be as clean as possible to reduce the difference between the experiment and the simulation. A high-pressure inert gas (e.g., nitrogen or argon) can produce cutting edges with higher quality, oxidation-free finishes, no reaction with the molten metal, and minimal heat generation. An advanced system with fully automized alignment and laser machining can reduce manufacturing errors. With the improved manufacturing system, we can reduce the gap between the simulation and experimental results. Furthermore, a polytetrafluoroethylene coating can help seal the patterned structure and reduce friction between the patterned tubes.

Numerical homogenization is an efficient method for determining the effective material properties of composite structures [44], [45]. In numerical homogenization, it is assumed that the unit cell is locally repeated with very small microstructures compared with the overall macroscopic dimensions of the structure of interest [45]. When the pattern is applied at the microscale and repeated more densely, this method can be applied to model the effective stiffness of the proposed mechanism. The accuracy of the homogenization model increases as the number of patterns increases and the size of the patterns decreases. For future work, we will design the robot with micro/nano-sized patterns and apply homogenization modeling.

Finally, the proposed optimization method potentially increases the number of applications of this variable stiffness technology. We plan to apply the design optimization method and the stiffness control methodology to variable stiffness continuum backbones, catheters, and needles for use in MIS or microrobots.

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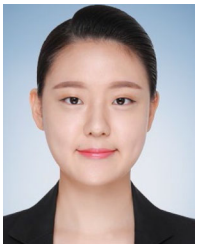

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